

ABIR DEOL

abir.deol2006@gmail.com · (672) 968-0839 · Edmonton, AB
abirdeol.tech · github.com/apollo-2006 · linkedin.com/in/abirdeol

EDUCATION

University of Alberta · Bachelor of Science, Computing Science

Expected 2028

TECHNICAL SKILLS

Languages: C, C++17, Rust, Python, Go, TypeScript, JavaScript, GLSL, SQL

Systems: Memory safety and sanitizers (ASan, UBSan, Miri), concurrency, memory management, distributed consensus, storage engines, GPU compute and SPIR-V, TCP and WebSockets

Tools: Linux, Git, GDB, CMake, Vulkan validation layers, LLVM/Clang, SDL2, FastAPI, React, SQLite

OPEN-SOURCE CONTRIBUTIONS

19 pull requests and 9 issues across 10 upstream repositories, 6 merged. Most began as a bug I found, reported, and then fixed myself. Write-ups at abirdeol.tech/work.

Khronos Group · Vulkan-ValidationLayers | C++, SPIR-V, GPU-AV **2 MERGED**

- Fixed a segfault in `vkCreateComputePipelines` that took down any compute shader whose `coopMatLoad` or `coopMatStore` used a stride computed at runtime: the GPU-AV instrumentation pass read the stride as if it were always a constant. Reported with a minimal shader, then wrote the fix and two regression tests (#13098 to #13104).
- Fixed an abort when GPU-AV decoded a shader offset that never named an instruction: an out-of-bounds shared-memory index makes the recovered word arbitrary, and decoding it underflowed a word count and indexed an empty operand list. Validated the offset before decode, letting the decoder assert the invariant instead (#13134 to #13150, through five review rounds with the maintainer).

ggml-org · llama.cpp and whisper.cpp | C++, GLSL, Vulkan, CI **3 MERGED**

- Found, with the validation layers, that both Vulkan `im2col` shaders declared a `buffer_reference` with no alignment, so every store was emitted as `Aligned 16` against a real 2 or 4 byte stride. One word per shader took 40 VUID violations to zero (#28960 to #28996).
- Fixed the `whisper.cpp` test suite aborting on every `GGML_BACKEND_DL=ON` build: no backend registers until something calls `ggml_backend_load_all()`, and the tests were the only callers that did not (#4030 to #4031). Documented the stream example's VAD timestamps and model-path failure mode (#4019).
- Open: a heap out-of-bounds read reachable from `whisper-cli --vad` on any user-supplied VAD model, found under AddressSanitizer with a crafted 94-byte model (#4064), plus a CI leg covering the build that broke above.

AMD · lemonade SDK and ROCm TheRock | C++, Python **1 MERGED**

- AMD GPUs on Linux were named by their raw KFD `gfx_target_version`, so an RX 9070 XT reported itself as `120001`. That field also keys the ROCm architecture lookup, so the readable name had to go beside it rather than replace it (#3592 to #3601).
- Traced embedding requests over 512 tokens failing with a 500 under an 8192 token context: an embedding model is non-causal, so the physical micro-batch is the real ceiling and had regressed to the default (#3591 to #3611).

Memory-safety fixes | *Hugging Face candle and tokenizers, espeak-ng, pdf-rs, Rust cargo*

- Undefined behaviour loading quantized GGML tensors, a byte slice cast to blocks with no length or alignment check; and a heap overflow in `Tensor::from_raw_buffer` reachable from safe code, where a seven-byte f32 buffer wrote past its allocation and returned `Ok` (candle #3963, #3965, found with Miri).
- A malformed BPE merge list producing invalid UTF-8 through `from_utf8_unchecked`; the fix deletes the unsafe block (tokenizers #2398). An unsigned subtraction reading before a heap string on ordinary voice selection, found under UBSan (espeak-ng #2530).

PROJECTS

oracle_of_delphi · Local Voice Assistant | Rust, C++, TypeScript, Three.js

- Built a fully local voice assistant as cooperating processes behind one executable: a real-time C++ audio layer for capture, VAD and barge-in, a Rust orchestrator running the agent loop, and a privileged actuator daemon. Boots offline with no GPU, model download or credentials, then swaps in real backends behind traits.
- Isolated all OS actuation in the privileged daemon, treating the language model as an untrusted planner: it recomputes the required capability from the operation itself, gates irreversible actions behind confirmation, and audits over a `SO_PEERCREDS` socket. 481 Rust tests and 933 C++ checks pass on Linux, macOS and Windows CI.

nexus_db · Embedded LSM Key-Value Store | C++17, Python/FastAPI, React

- Built a storage engine modeled on LevelDB: a skip-list memtable for $O(\log n)$ insertion and point lookup, a write-ahead log appending every operation before the memtable is modified, frozen memtables flushed to immutable SSTables, levelled compaction, Bloom filters to cut reads that touch disk for keys that are not there, and tombstone deletion reclaimed during compaction. Exposed over a ctypes FFI to Python/FastAPI and a React dashboard.

nexus_cluster · Distributed Consensus Engine | Python, Raft, TCP

- Implemented Raft from the paper with no consensus library: leader election, log replication and term management over a custom TCP transport, durable term and vote so a restarted node cannot vote twice, log compaction by snapshot, and a read-index path for linearizable reads that do not append an entry. A simulation harness checks the paper's safety properties after every step as nodes fail and rejoin.

nano_match · Limit Order Book and Matching Engine | C++

- Built a price-time priority matching engine covering the full order lifecycle. Eliminated all heap allocation after startup with a custom pool, reaching 12M requests/s at a 50ns median on one core.

custom_mem_alloc · Thread-Safe Memory Allocator | C, pthreads

- Wrote a `malloc` replacement over a single `mmap` region: first-fit search, block splitting, neighbour coalescing on free, 8-byte alignment and mutex locking, benchmarked against `glibc`.

Additional: `nexus_editor` (CRDT editor, Go) · `photon_tracer` (CPU path tracer, 1080p at 50 spp in half a second on 32 threads) · `neon_vm` (bytecode VM) · `cpu_rasterizer` and `rasterizer_engine` (software 3D pipelines, JS and C++/SDL2) · `radiant_slice` (FPS systems, Unreal Engine 5) · `thermal_monitor`, `terminal_dashboard`, `valo_scout`, `cal-cli`. 29 public repositories, 10 with live demos.

WRITING AND ACTIVITIES

Engineering post-mortems at abirdeol.tech/research: five write-ups of bugs that cost me real hours, each covering the symptom, the chase, the real cause, and what I changed. **Mixed Martial Arts**: two-time national champion, 2020 to 2024, training six days a week alongside full-time coursework.